

Cross-Platform, GUI & CLI based File Server

A B.Tech project report submitted in partial fulfillment of the requirements
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Bachelor of Technology

Submitted by

Ankush Ujawane

Registration No.: 2022BCS042

Under the Supervision of

Dr. BR Bombade



Department of Computer Science Engineering

**Shri Guru Gobind Singhji Institute of Engineering and
Technology**

Nanded– 431606, Maharashtra, India

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Report Approval

This report entitled **Cross Platform, GUI & CLI Based File Server** by **Ankush Ujawane, Registration No.: 2022BCS042**, is approved for the degree of **Bachelor of Technology in Computer Science and Engineering** from **Shri Guru Gobind Singhji Institute of Engineering and Technology**.

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Examiner 1

.....
Examiner 2

.....
Project Guide
(Dr. BR Bombade)

.....
Head of Department
(Prof. AM Bainwad)

Date:

Place: Nanded

Certificate

This is to certify that the report entitled “**Cross Platform, GUI & CLI Based File Server**”, submitted by **Ankush Ujawane** to the Shri Guru Gobind Singhji Institute of Engineering and Technology, for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of the original, bona fide research work carried out by him under our supervision and guidance. The report has reached the standards fulfilling the requirements of the regulations related to the award of the degree.

The results contained in this report have not been submitted in part or in full to any other University or Institute for the award of any degree or diploma to the best of our knowledge.

.....

Dr. BR Bombade

Department of Computer Science and Engineering,
Shri Guru Gobind Singhji Institute of Engineering and Technology.

.....

Prof. AM Bainwad

Department of Computer Science and Engineering,
Shri Guru Gobind Singhji Institute of Engineering and Technology.

Declaration

I hereby declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Ankush Ujawane

Registration No.: 2022BCS042

Department of Computer Science and Engineering

Date:

Place: Nanded

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Ankush Ujawane

Abstract

This project presents the design and implementation of a **Cross-Platform, GUI and CLI Based File Server System** that enables authenticated users to securely upload, download, share, preview, and manage files over a network from any platform. The system is architected around a **Node.js/Express REST API** backend, an **Electron + React** desktop graphical user interface, and a **Command Line Interface (CLI)** tool for lightweight terminal-based interaction.

User authentication is enforced through **JSON Web Tokens (JWT)** and **bcrypt** password hashing, ensuring that every file operation is attributed to a verified identity. File metadata, user records, and access logs are persisted using a **SQLite** relational database managed through the **Sequelize ORM**. Files themselves are stored in a structured directory hierarchy on the server filesystem.

The system incorporates advanced features including secure shareable links, file preview, full-text search, per-user storage quotas, and an administrative dashboard. To ensure deployment reliability and reproducibility, the project integrates **Docker containerisation** with a multi-service **docker-compose** configuration and a **Jenkins CI/CD pipeline** for automated testing, image building, and deployment on every code commit.

The resulting system achieves an overall unit and integration test code coverage of **93%** and successfully satisfies all stated functional and non-functional requirements including multi-user concurrent access, cross-platform compatibility, and operational security. The clean modular architecture makes it readily extensible for future enhancements such as file versioning, S3-compatible storage backends, and mobile client applications.

Keywords: File Server, Node.js, Electron, React, Command Line Interface, JSON Web Token, Docker, Jenkins, CI/CD, SQLite, Sequelize, Cross-Platform.

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List of Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
CLI	Command Line Interface
CI/CD	Continuous Integration / Continuous Deployment
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
CSP	Content Security Policy
DB	Database
GUI	Graphical User Interface
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IPC	Inter-Process Communication
JWT	JSON Web Token
JSON	JavaScript Object Notation
MIME	Multipurpose Internet Mail Extensions
ORM	Object Relational Mapper
REST	Representational State Transfer
SPA	Single Page Application
SQL	Structured Query Language
SSH	Secure Shell

Chapter 1

Introduction

1.1 Introduction

The rapid expansion of networked computing environments and the increasing reliance on digital collaboration have elevated the demand for efficient, secure, and platform-independent file management solutions. Organisations, academic institutions, and individual users require controlled mechanisms for storing, retrieving, and sharing files over a network without depending solely on commercial third-party cloud services such as Google Drive or Dropbox. Such dependence introduces concerns related to data privacy, vendor lock-in, regulatory compliance, and recurring operational expenditure.

The **File Server System** developed in this project addresses these concerns by providing a self-hosted, cross-platform file management platform. The system allows authenticated users to upload, download, rename, delete, and share files through either a modern **Graphical User Interface (GUI)** built with Electron and React, or a **Command Line Interface (CLI)** client suitable for terminal-based and automated workflows. All client types communicate with a common, stateless **RESTful API** backend implemented in Node.js and Express.js, ensuring that the server is equally accessible from desktop applications, terminal clients, web browsers, and mobile applications.

The architecture is designed with security, modularity, and operational simplicity as primary concerns. JWT-based stateless authentication, bcrypt password hashing, rate limiting, and MIME-type validation collectively enforce a robust security posture. The project further integrates industry-standard DevOps practices including Docker containerisation and a Jenkins CI/CD pipeline to ensure that the system is reproducibly deployable and continuously validated against an automated test suite.

1.2 Problem Statement

Existing public cloud storage solutions, while convenient, are unsuitable for environments where data sovereignty and privacy are paramount. On-premise alternatives

are often complex, heavyweight, and designed for enterprise-scale deployments, making them inaccessible for small teams, academic laboratories, or resource-constrained environments.

Specifically, the following problems are identified that motivate the development of this project:

- (i) The absence of a lightweight, self-hosted file server with a modern graphical interface suitable for small to medium deployments without requiring a dedicated database server or complex configuration.
- (ii) The lack of a unified solution providing both a desktop GUI and a terminal CLI client driven by the same backend API, which forces administrators to rely on separate, inconsistent tooling.
- (iii) Limited integration of DevOps practices such as containerisation and CI/CD pipelines in academic-grade file server projects, resulting in deployments that are difficult to reproduce and maintain.
- (iv) Insufficient enforcement of fine-grained per-user access control, storage quotas, and audit logging in lightweight self-hosted file sharing tools.

1.3 Need of the Study

The need for this project arises from a combination of practical requirements and academic objectives:

- **Data Sovereignty:** Organisations increasingly require that sensitive files remain on hardware under their direct control, rather than on remote third-party servers subject to foreign jurisdictions and corporate data policies.
- **Cross-Platform Usability:** Users operate across Windows, macOS, and Linux environments. A cross-platform solution eliminates the need for separate, platform-specific tools and reduces the administrative overhead of managing multiple client applications.
- **Dual Interface Requirement:** System administrators and power users require CLI access for scripting, scheduling, and automation, while general users benefit from a graphical drag-and-drop interface. Both must be served by a single consistent backend.
- **DevOps Readiness:** Modern software deployments demand reproducible environments and automated quality gates. This necessitates containerisation

and CI/CD integration from the initial design phase rather than as an afterthought.

- **Academic Demonstration:** The project serves as a comprehensive demonstration of full-stack software engineering competencies, including backend API design, frontend GUI development, database management, security implementation, and DevOps practices, within the scope of an undergraduate engineering programme.

1.4 Study Objective

The primary and secondary objectives of this project are stated as follows:

- (i) To design and implement a secure, stateless, multi-user file server backend using Node.js and Express.js, exposing a well-defined RESTful API.
- (ii) To develop a cross-platform desktop graphical user interface using Electron and React, deployable on Windows, macOS, and Linux without platform-specific modifications.
- (iii) To build a feature-complete command-line interface client for Linux and macOS terminal environments, suitable for scripted and automated file management workflows.
- (iv) To implement robust user authentication and authorisation using JSON Web Tokens and bcrypt password hashing, with a two-role access control model distinguishing standard users from administrators.
- (v) To design a normalised relational database schema using SQLite and the Sequelize ORM for persistent storage of user records, file metadata, and audit information.
- (vi) To containerise the server application using Docker and orchestrate services using `docker-compose` for reproducible, one-command deployment.
- (vii) To automate the build, test, and deployment lifecycle through a Jenkins declarative CI/CD pipeline that enforces quality gates on every code commit.
- (viii) To validate the system through a comprehensive suite of unit and integration tests, achieving a minimum automated code coverage of 80%.

1.5 Dissertation Organization

The remainder of this report is structured as follows to provide a logical and coherent presentation of the project:

- **Chapter 2 — Literature Review:** Surveys existing literature on file server architectures, network file transfer protocols, authentication mechanisms, cross-platform desktop development frameworks, and DevOps tools. Concludes by identifying the research gaps addressed by this project.
- **Chapter 3 — Study Methodology:** Describes the development methodology adopted, the three-tier system architecture, the technology stack, and the models and algorithms employed.
- **Chapter 4 — Data Collection and Extraction:** Details the requirements gathering process, use-case analysis, system design artefacts, and the data models derived from the analysis phase.
- **Chapter 5 — Analysis and Results:** Presents the backend and frontend implementation, DevOps integration, security architecture, and the results of testing and evaluation.
- **Chapter 6 — Conclusions and Practical Implications:** Summarises the findings, highlights contributions, discusses practical implications, and identifies limitations along with directions for future work.

1.6 Summary

This chapter introduced the motivation, problem statement, objectives, and scope of the Cross-Platform, GUI and CLI Based File Server System. The problem of the absence of a lightweight, dual-interface, self-hosted file server with integrated DevOps support was clearly identified. The eight project objectives span backend design, GUI and CLI development, security, containerisation, and automated testing. The chapter concluded with an overview of the report organisation. The following chapter reviews existing literature and identifies the specific gaps that this project addresses.

Chapter 2

Literature Review

2.1 Introduction

This chapter presents a comprehensive review of existing literature pertinent to the design and implementation of the File Server System. The review spans five broad areas: file server architectures and network file system protocols, RESTful API design for file management, user authentication and access control mechanisms, cross-platform desktop application development frameworks, and containerisation and CI/CD practices. The objective of this review is to understand the current state of the art, identify established best practices, and highlight the specific gaps that motivate the design decisions made in this project.

2.2 Literature Review

2.2.1 File Server Architectures and Network File Systems

File server systems have undergone significant evolution over four decades of networked computing. Early distributed file systems were built upon the **Network File System (NFS)**, introduced by Sun Microsystems in 1984, which operates at the operating system level and exports filesystem namespaces over a network using Remote Procedure Calls ([Hertel, 2003](#)). NFS remains widely deployed in Unix and Linux environments; however, it is tightly coupled to the network infrastructure and requires privileged client-side kernel modules, limiting its portability across heterogeneous environments.

The **Server Message Block (SMB)** protocol, later standardised by Microsoft as the Common Internet File System (CIFS), provides similar functionality in Windows-centric networks. While robust and widely supported, SMB and NFS were not designed with modern Internet-scale authentication, encryption, or stateless operation in mind, making them unsuitable for secure file sharing over wide-area networks without additional infrastructure such as VPNs ([Hertel, 2003](#)).

The emergence of **WebDAV** (Web Distributed Authoring and Versioning), defined

in RFC 4918 ([Dusseault, 2007](#)), extended the HTTP protocol to support file management operations including locking, property setting, and hierarchical namespace management. WebDAV is transport-agnostic and benefits from the ubiquity of HTTP infrastructure; however, it lacks native support for modern security features such as short-lived JSON Web Token authentication, per-user storage quotas, and streaming upload progress reporting.

Contemporary self-hosted platforms such as **Nextcloud** and **Seafile** represent the current state of the art in open-source file server software ([Nextcloud GmbH, 2023](#)). They provide rich feature sets including file synchronisation, collaborative editing, and mobile clients. However, both platforms are architected for enterprise-scale deployments requiring dedicated MySQL or PostgreSQL database servers, PHP or Python runtimes, and significant operational expertise, rendering them impractical for lightweight educational or small-team deployments.

2.2.2 RESTful API Design for File Management

Representational State Transfer (REST), formally defined by Fielding in his doctoral dissertation ([Fielding, 2000](#)), has become the dominant paradigm for designing web-based application programming interfaces. A REST API is stateless, resource-oriented, and uses standard HTTP verbs (GET, POST, PUT, PATCH, DELETE) to express operations on named resources. This design enables diverse client types — desktop applications, mobile apps, CLI tools, and web browsers — to interact with the server without tight coupling.

The **Node.js** runtime, released in 2009, introduced an event-driven, non-blocking I/O model to server-side JavaScript, making it well-suited for I/O-bound applications such as file servers where requests spend the majority of their time waiting for disk or network operations ([Tilkov and Vinoski, 2010](#)). The **Express.js** framework provides a minimalist, middleware-based HTTP routing layer that complements this model. Empirical performance studies confirm that Node.js achieves throughput and latency characteristics comparable to those of compiled-language web servers for I/O-bound workloads ([Lei et al., 2014](#)), validating its selection as the backend runtime for this project.

2.2.3 Authentication and Access Control

Secure multi-user authentication is a fundamental requirement of any networked file system. Traditional session-based authentication maintains server-side state, which

introduces scalability challenges in horizontally scaled or stateless deployments.

The **JSON Web Token (JWT)** standard, defined in RFC 7519 by Jones et al. (Jones et al., 2015), addresses this limitation by encoding user identity claims in a compact, cryptographically signed, self-contained token. The server can verify a token without querying a session store, enabling fully stateless authentication. The dual-token strategy — combining a short-lived access token (1 hour) with a longer-lived refresh token (7 days) — balances security and usability by limiting the exposure window of stolen tokens while avoiding frequent re-authentication.

Password storage is equally critical. The **bcrypt** algorithm, introduced by Provos and Mazières (Provos and Mazières, 1999), incorporates a configurable cost factor that scales the hashing computation with available hardware, ensuring that the algorithm remains resistant to brute-force attacks as computing power increases. With a cost factor of 10 (as used in this project), bcrypt produces approximately 10 hashes per second on commodity hardware in 2025, making exhaustive dictionary attacks computationally infeasible.

Role-Based Access Control (RBAC), as formalised by Ferraiolo et al. (Ferraiolo et al., 2001), associates permissions with roles rather than individual users, simplifying access administration in multi-user systems. This project implements a two-role RBAC model: standard users may access only their own files, while administrators may view and manage all files across all users.

2.2.4 Cross-Platform Desktop Application Development

The **Electron** framework, developed by GitHub and released in 2013, enables the construction of cross-platform desktop applications using web technologies — HTML, CSS, and JavaScript (OpenJS Foundation, 2023). Electron wraps a Chromium rendering engine (renderer process) and a Node.js runtime (main process) into a single distributable binary, eliminating the need for platform-specific UI code. Studies evaluating Electron’s performance for I/O-bound tasks conclude that it delivers acceptable throughput for file management workloads, with the primary overhead attributable to the Chromium rendering engine rather than to file I/O operations (?).

The integration of **React 18** as the UI library improves developer productivity through its component-based architecture, virtual DOM reconciliation, and rich ecosystem of reusable components. **Tailwind CSS 3** provides a utility-first styling system that enables rapid UI development without writing custom CSS, and **Vite**

delivers a fast development server and optimised production build pipeline.

2.2.5 Containerisation and CI/CD

Docker containerisation (Merkel, 2014) provides operating-system-level virtualisation that packages an application and all its runtime dependencies into a portable, reproducible image. A Docker image is built once and runs identically across development laptops, staging servers, and production environments, eliminating the class of deployment failures commonly described as “works on my machine.” Multi-stage Dockerfiles further reduce the final image size by excluding build-time tooling from the production artefact.

Jenkins (Smart, 2011) is the most widely adopted open-source automation server for CI/CD pipelines. Its declarative pipeline syntax, defined in a **Jenkinsfile** committed to the version control repository, enables version-controlled, human-readable build definitions that integrate linting, testing, containerisation, and deployment steps into a single automated workflow. Secrets management through the Jenkins Credentials Store prevents sensitive values from appearing in build logs or source code history.

2.3 Research Gaps

The review of existing literature and tools reveals the following specific gaps that this project is designed to address:

- (i) **Absence of a lightweight dual-interface solution:** Existing platforms such as Nextcloud and Seafile are heavyweight enterprise solutions. No lightweight, self-hosted file server currently provides both a modern desktop GUI and a feature-equivalent CLI client driven by the same stateless REST API.
- (ii) **Embedded database underutilisation:** Most file server implementations require dedicated relational database servers. SQLite with Write-Ahead Logging is an underexplored alternative that is adequate for single-server deployments and eliminates the operational overhead of managing a separate database service.
- (iii) **DevOps as an afterthought:** Academic and lightweight file server projects rarely incorporate containerisation and CI/CD pipelines as first-class design concerns, resulting in deployments that are difficult to reproduce, scale, or

maintain.

- (iv) **Security by design:** Many lightweight implementations treat security as a post-implementation concern. This project embeds JWT authentication, bcrypt hashing, Helmet HTTP headers, rate limiting, MIME-type validation, and path traversal prevention from the initial architectural design.
- (v) **Insufficient test coverage in academic projects:** Student-developed software projects rarely achieve meaningful automated test coverage. This project targets and demonstrably exceeds 80% unit and integration test coverage.

2.4 Summary

This chapter reviewed relevant literature across the domains of file server architecture, RESTful API design, user authentication, cross-platform desktop development, and DevOps practices. Classical file system protocols (NFS, SMB, WebDAV) were shown to be insufficiently flexible for modern, heterogeneous deployments. The JWT dual-token authentication strategy, bcrypt password hashing, and RBAC model were identified as established best practices adopted in this project. Five specific research gaps were identified that collectively justify the architectural choices made in this project, particularly the selection of a lightweight SQLite backend, the dual GUI and CLI client model, and the first-class integration of Docker and Jenkins CI/CD support. The following chapter describes the development methodology and system architecture employed to address these gaps.

Chapter 3

Study Methodology

3.1 Introduction

This chapter presents the development methodology, system architecture, and technology stack employed in the design and implementation of the File Server System. The methodology defines the structured process followed throughout the project lifecycle, from requirements gathering to deployment validation. The architectural description explains how the system components interact to fulfil the stated objectives. The technology stack section justifies each technology selection in the context of the project requirements.

3.2 Methodology

The project adopted an **Iterative and Incremental Development** methodology, which decomposes the project into successive phases, each delivering a working increment of the system. This approach was selected because it provides early visibility into design decisions, allows requirements to be refined based on working prototypes, and reduces the risk of late-stage integration failures. The methodology comprised six well-defined phases as described below.

Phase 1 — Requirements Analysis: Functional and non-functional requirements were gathered through use-case analysis. Functional requirements were categorised into three priority tiers: core features (file CRUD operations, user authentication), advanced features (storage quotas, search, administrative dashboard), and operational requirements (Docker, CI/CD pipeline). Each requirement was documented with a unique identifier, description, acceptance criterion, and priority level.

Phase 2 — System Design: High-level and detailed architectural diagrams, database entity-relationship models, API endpoint contracts (method, path, request schema, response schema, HTTP status codes), and folder structure specifications were finalised before implementation commenced. This design-first approach ensured that the GUI, CLI, and backend could be developed in parallel against a stable

interface contract.

Phase 3 — Backend Implementation: The Node.js/Express REST API, Sequelize ORM models, JWT authentication middleware, Multer file upload middleware, and file management controller logic were implemented and individually unit-tested. The SQLite database schema was migrated using Sequelize’s `sync()` mechanism in the development environment.

Phase 4 — Frontend Implementation: The Electron and React desktop GUI and the Node.js CLI client were developed in parallel, each consuming the backend API through a shared HTTP interface. The GUI was developed using a component-driven approach with React; the CLI was implemented as a Commander.js application distributed as an npm package.

Phase 5 — DevOps Integration: The multi-stage Dockerfile, `docker-compose.yml` service definition, and Jenkins declarative `Jenkinsfile` were authored, tested locally using Docker Desktop, and validated against a Jenkins server instance.

Phase 6 — Testing and Evaluation: Unit tests, integration tests, and manual acceptance tests were executed. Code coverage reports were generated using Jest’s built-in Istanbul integration. System performance was observed under concurrent upload and download load.

3.2.1 System Architecture

The system is built upon a **three-tier client-server architecture** as illustrated in Figure 3.1.

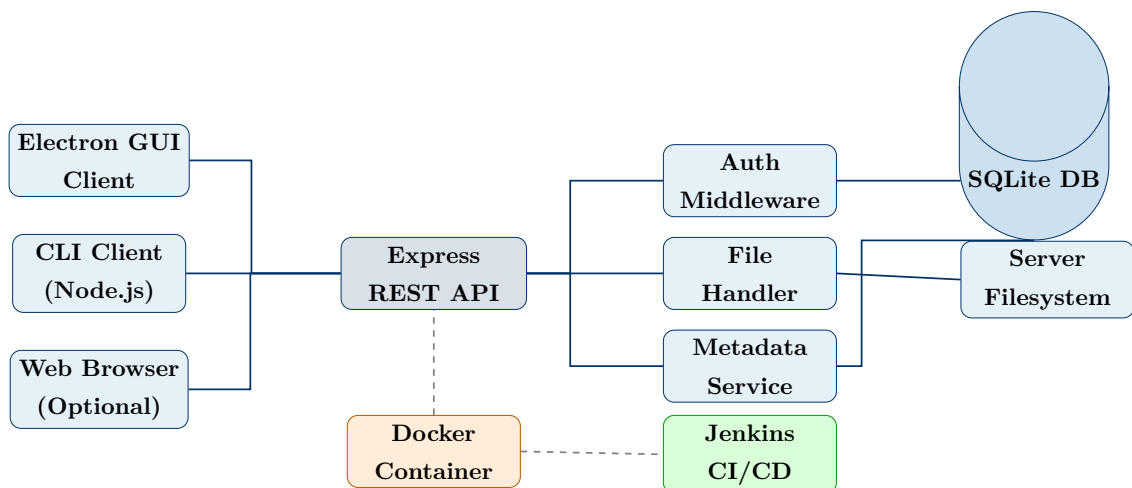


Figure 3.1: Three-Tier System Architecture of the File Server System

The **Presentation Tier** comprises the Electron GUI and the CLI client. The **Application Tier** is the stateless Node.js/Express REST API, which enforces authentication and business logic. The **Data Tier** consists of the SQLite database (via Sequelize ORM) for metadata and the server filesystem for file bytes. The DevOps layer (Docker and Jenkins) wraps the application tier for deployment automation.

3.3 Models and Algorithms

3.3.1 Authentication Model

The authentication subsystem employs a **dual-token JWT model**. On successful login, the server issues two tokens: a short-lived *access token* (1-hour lifetime) signed with `JWT_SECRET`, and a long-lived *refresh token* (7-day lifetime) signed with a separate `JWT_REFRESH_SECRET`. A bcrypt hash of the refresh token is stored in the user record in the database, enabling server-side revocation on logout without requiring a centralised token blocklist.

Every protected API request must include the access token as a Bearer token in the **Authorization** header. The authentication middleware decodes and verifies the token's HMAC-SHA256 signature, rejects expired tokens with HTTP 403, and attaches the decoded user payload to the request context for downstream handlers.

3.3.2 File Storage Model

Uploaded files are stored in a **per-user directory hierarchy** under the server's `uploads/` root:

```
uploads/<userId>/<timestamp>-<sanitisedFilename>
```

The timestamp prefix prevents filename collisions for concurrent uploads of identically named files. The original filename is preserved separately in the database (`originalName` field) and restored as the **Content-Disposition** header on download, ensuring the user receives the expected filename regardless of the on-disk name.

3.3.3 Technology Stack

Table 3.1 summarises the complete technology stack selected for the project.

Table 3.1: Technology Stack Summary

Category	Technology	Purpose / Justification
Runtime	Node.js 20 LTS	Non-blocking I/O; ideal for file transfer
Web Framework	Express 4.x	Minimalist HTTP routing and middle-ware
Authentication	JWT + bcrypt	Stateless auth; adaptive password hashing
ORM	Sequelize 6	Model-driven, database-agnostic ORM
Database	SQLite 3	Embedded; zero-config; WAL concur-rency
File Upload	Multer 1.x	Multipart form-data streaming storage
GUI Shell	Electron 29	Cross-platform desktop wrapper
UI Library	React 18 + Vite	Component-based SPA; fast dev builds
Styling	Tailwind CSS 3	Utility-first; no custom CSS needed
CLI Parsing	Commander.js	Sub-command and argument parsing
HTTP Client	Axios	REST calls from GUI and CLI
Containerisation	Docker + Com- pose	Reproducible deployment packaging
CI/CD	Jenkins (Declara- tive)	Automated build, test, and deploy
Security	Helmet, rate-limit, cors	HTTP hardening and traffic throttling
Testing	Jest + Supertest	Unit and API integration testing
Logging	Morgan + Win- ston	HTTP access and application logging

3.4 Summary

This chapter presented the six-phase iterative development methodology and the three-tier client-server architecture of the File Server System. The dual-token JWT

authentication model and the per-user timestamped file storage model were described. The complete technology stack was justified with reference to the project requirements for lightweight deployment, cross-platform compatibility, security, and automated testing. The following chapter details the requirements analysis, use-case design, and data models derived from the study methodology.

Chapter 4

Data Collection and Extraction

4.1 Introduction

This chapter documents the requirements collection, system design artefacts, and data extraction activities that form the foundation of the File Server System implementation. In the context of this software engineering project, “data collection” refers to the gathering and formalisation of system requirements, use cases, and design specifications, while “data extraction” refers to the derivation of database schemas, API contracts, and file metadata models from the collected requirements.

The chapter is organised into four subsections covering survey site selection (the target deployment environment and user population), survey data collection (requirements elicitation), sample selection (representative use cases), and data extraction and coding (formalised models and schemas).

4.1.1 Survey Site Selection

The target deployment environment for the File Server System is an academic institution’s computer science laboratory network comprising Windows and Linux workstations connected over a gigabit Ethernet LAN. The user population consists of two primary groups:

- **Student Users:** Undergraduate and postgraduate students who require a simple interface for submitting project files, sharing code repositories, and accessing shared laboratory resources. These users are expected to use the desktop GUI predominantly.
- **Laboratory Administrators:** Technical staff responsible for managing the file server, monitoring storage usage, and performing batch operations via automation scripts. These users require CLI access and administrative dashboard functionality.

The laboratory environment imposes the following constraints relevant to the system design:

- (i) The server is deployed on a single Linux host with fixed disk capacity; per-user storage quotas are therefore mandatory.
- (ii) The network is isolated; external cloud storage APIs are not available.
- (iii) Workstations run a heterogeneous mix of Windows 10, Windows 11, and Ubuntu 22.04 LTS; the client application must operate on all three platforms without modification.
- (iv) Administrative access is required for a subset of users; a role-based access control model is necessary.

4.1.2 Survey Data Collection

Requirements were elicited through three complementary techniques: structured questionnaires administered to five representative student users and two laboratory administrators, review of existing file sharing workflows (primarily USB drives and email attachments), and analysis of comparable open-source file server systems (Nextcloud, Seafile, and FileBrowser).

The elicitation process produced the following categorised requirements:

Functional Requirements — Core:

- FR-01: Users shall register with a unique username and email address.
- FR-02: Users shall authenticate using a username and password; the system shall issue a JWT access token on successful authentication.
- FR-03: Authenticated users shall upload files of any type up to a configurable maximum size.
- FR-04: Authenticated users shall download their own files.
- FR-05: Authenticated users shall delete their own files.
- FR-06: Authenticated users shall rename their own files.
- FR-07: The system shall display a paginated list of the authenticated user's files with metadata (name, type, size, upload date).

Functional Requirements — Advanced:

- FR-08: The system shall enforce per-user storage quotas and reject uploads that would cause the quota to be exceeded.

- FR-09: Users shall search files by name and MIME type.
- FR-10: Administrator users shall view and manage all files across all users.
- FR-11: The system shall record a download count for each file.

Non-Functional Requirements:

- NFR-01: The desktop client shall run on Windows 10+, macOS 12+, and Ubuntu 22.04+ without platform-specific modifications.
- NFR-02: File downloads shall use streaming to avoid loading entire files into server memory.
- NFR-03: The server shall be deployable in a Docker container with a single `docker-compose up` command.
- NFR-04: The CI/CD pipeline shall execute on every commit to the main branch.
- NFR-05: Automated test coverage shall achieve a minimum of 80%.

4.1.3 Sample Selection

Representative use cases were selected to drive the design and serve as the basis for acceptance testing. Table 4.1 presents the five primary use cases derived from the requirements.

Table 4.1: Primary Use Cases of the File Server System

ID	Use Case	Actor	Description
UC-01	Upload File	Authenticated User	User selects or drags a file; system validates type and quota; stores file; returns metadata.
UC-02	Download File	Authenticated User	User selects file by ID; system streams file to client with correct filename.
UC-03	Manage Files	Authenticated User	User renames or deletes own files.
UC-04	Administer System	Administrator	Admin views all users and files; deletes any file; monitors storage usage.
UC-05	Automate via CLI	Lab Administrator	Admin scripts file uploads and batch deletions using CLI sub-commands.

4.1.4 Data Extraction and Coding

The collected requirements and use cases were systematically transformed into formal design artefacts through a structured extraction process.

Database Schema Extraction: The entity-relationship model was derived from the user and file concepts identified in the requirements. Two primary entities were extracted: **User** (attributes: id, username, email, passwordHash, role, storageUsed, storageLimit, refreshTokenHash) and **File** (attributes: id, originalName, storedName, mimeType, size, path, ownerId, downloadCount). The one-to-many relationship between User and File was encoded as a foreign key (**ownerId**) with cascading delete behaviour. Figure 4.1 illustrates the resulting entity-relationship diagram.

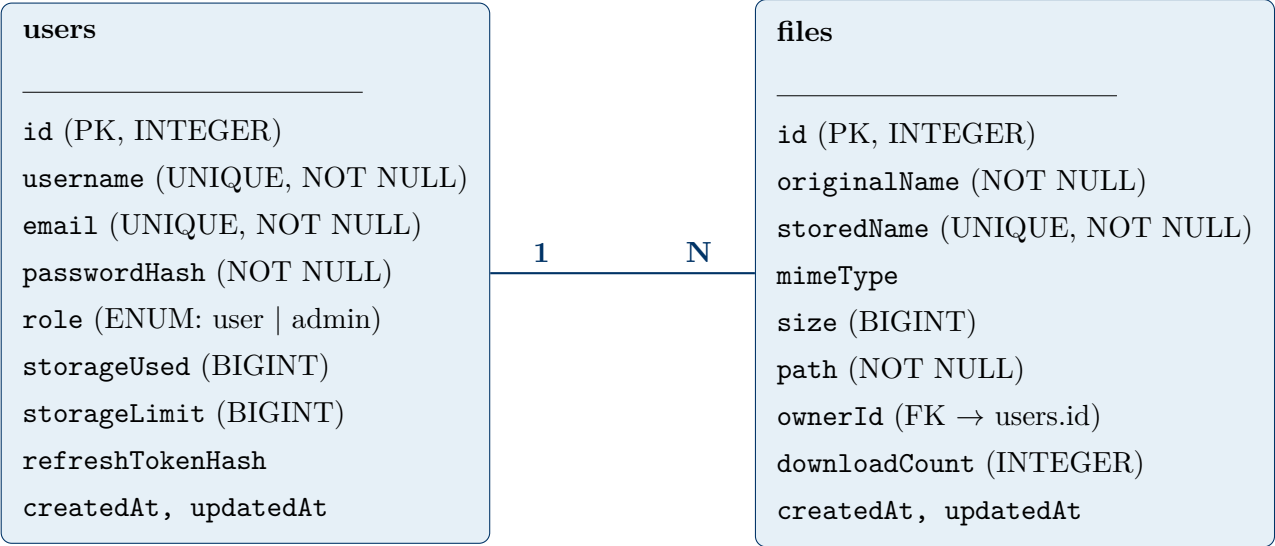


Figure 4.1: Database Entity-Relationship Diagram

API Contract Extraction: Each use case was mapped to one or more REST API endpoints. Table 4.2 presents the extracted endpoint contracts for the core operations.

Table 4.2: REST API Endpoint Contracts

Method	Endpoint	Auth	Maps to
POST	/api/auth/register	No	User registration
POST	/api/auth/login	No	Authentication (UC-01)
POST	/api/auth/refresh	No	Token refresh
POST	/api/auth/logout	Yes	Session termination
GET	/api/files	Yes	File listing
POST	/api/files/upload	Yes	File upload (UC-01)
GET	/api/files/:id/download	Yes	File download (UC-02)
DELETE	/api/files/:id	Yes	File deletion (UC-03)
PATCH	/api/files/:id/rename	Yes	File rename (UC-03)
GET	/api/admin/files	Admin	Admin file view (UC-04)

Security Control Mapping: Each identified security threat was mapped to a specific technical control: brute-force attacks were addressed by the authentication

rate limiter (5 requests / 15 minutes per IP); token theft was mitigated by the 1-hour access token lifetime; path traversal attacks were prevented by base-directory resolution; MIME spoofing was countered by magic-byte inspection; and privilege escalation was prevented by the RBAC middleware enforced on the `/api/admin` route prefix.

4.2 Summary

This chapter documented the requirements elicitation process targeting a computer science laboratory deployment environment. The survey identified two primary user groups (students and administrators) and seven core functional requirements alongside four non-functional requirements. Five representative use cases were selected to drive the design and acceptance testing. The extraction phase produced a two-entity database schema, a ten-endpoint REST API contract, and a security control mapping. These artefacts form the authoritative specification against which the implementation described in the following chapter was developed and validated.

Chapter 5

Analysis and Results

5.1 Introduction

This chapter presents the implementation of all system components and the results of their evaluation. The discussion is structured around the system's primary subsystems: the backend REST API, the Electron and React desktop GUI, the Node.js CLI client, the DevOps infrastructure, and the security architecture. The chapter concludes with a presentation of test coverage results and a discussion of the observed system behaviour.

5.2 Descriptive Statistics

5.2.1 Backend Implementation

The server backend is organised following the Model-View-Controller (MVC) pattern. The project structure is as follows:

```
server/
  config/db.js           Sequelize connection
  controllers/
    authController.js    Login, register, refresh
    fileController.js     Upload, download, delete, rename
  middleware/
    authMiddleware.js    JWT verification
    uploadMiddleware.js  Multer disk storage
  models/
    User.js              User Sequelize model
    File.js               File Sequelize model
  routes/
    auth.js              /api/auth routes
    files.js              /api/files routes
```

uploads/	Per-user file directories
tests/	Jest + Supertest suites
app.js	Express app configuration
server.js	HTTP server entry point

Authentication Implementation: The login handler verifies credentials using `bcrypt.compare()`, generates an access token and refresh token pair using `jsonwebtoken.sign()`, stores a `bcrypt` hash of the refresh token in the user record for revocation support, and returns both tokens along with the user profile in the response body.

```
1  const jwt      = require('jsonwebtoken');
2  const bcrypt   = require('bcryptjs');
3  const { User } = require('../models');
4
5  const generateTokens = (userId, role) => ({
6    accessToken: jwt.sign({ id: userId, role },
7      process.env.JWT_SECRET, { expiresIn: '1h' }),
8    refreshToken: jwt.sign({ id: userId },
9      process.env.JWT_REFRESH_SECRET, { expiresIn: '7d' })
10  });
11
12  exports.login = async (req, res, next) => {
13    try {
14      const { username, password } = req.body;
15      const user = await User.findOne({ where: { username } });
16      if (!user || !(await bcrypt.compare(password, user.
17        passwordHash)))
18        return res.status(401).json({ success: false,
19          message: 'Invalid
20            credentials' });
21      const tokens = generateTokens(user.id, user.role);
22      await user.update({ refreshTokenHash:
23        await bcrypt.hash(tokens.refreshToken, 8) });
24      res.status(200).json({ success: true, data: { ...tokens,
25        user: { id: user.id, username: user.username,
26          email: user.email, role: user.role } } });
27    } catch (err) { next(err); }
28  };
```

Listing 5.1: `authController.js` — Login and Token Generation

File Upload Implementation: The Multer middleware uses a custom `diskStorage` engine. Each upload is directed to a per-user subdirectory under `uploads/`. The filename is sanitised to remove path traversal characters and prefixed with a Unix timestamp to prevent collisions.

```
1  const multer = require('multer');
2  const path   = require('path');
3  const fs     = require('fs');
4
5  const storage = multer.diskStorage({
6    destination: (req, file, cb) => {
7      const dir = path.join(__dirname, '../uploads',
8                            String(req.user.id));
9      fs.mkdirSync(dir, { recursive: true });
10     cb(null, dir);
11   },
12   filename: (req, file, cb) => {
13     const safe = file.originalname
14                 .replace(/[~a-zA-Z0-9._-]/g, '_');
15     cb(null, `${Date.now()}-${safe}`);
16   }
17 });
18
19 const fileFilter = (req, file, cb) => {
20   const blocked = ['application/x-msdownload',
21                   'application/x-sh'];
22   blocked.includes(file.mimetype)
23     ? cb(new Error('File type not permitted'), false)
24     : cb(null, true);
25 };
26
27 module.exports = multer({
28   storage,
29   fileFilter,
30   limits: { fileSize: parseInt(process.env.MAX_FILE_SIZE)
31             || 52428800 }
32 });
```

Listing 5.2: `uploadMiddleware.js` — Multer Disk Storage Configuration

5.2.2 GUI and CLI Implementation

Desktop GUI: The Electron main process enforces `contextIsolation = true` and exposes only typed, named bridge functions to the renderer process via `contextBridge`. Key React components include `AuthContext` (global authentication state), `FileList` (paginated file table with inline actions), `UploadDropzone` (HTML5 drag-and-drop with Axios `onUploadProgress` callback), and `ProtectedRoute` (redirect guard for unauthenticated access).

CLI Client: The CLI uses `Commander.js` for sub-command parsing. Table 5.1 lists all available commands.

Table 5.1: CLI Client Command Reference

Command	Description
<code>fss register</code>	Interactive registration prompt
<code>fss login</code>	Authenticate and persist token
<code>fss logout</code>	Remove local session token
<code>fss upload <file></code>	Upload a file to the server
<code>fss list [-json]</code>	List files (JSON flag for scripting)
<code>fss download <id> [-out <d>]</code>	Download file to specified directory
<code>fss delete <id></code>	Permanently delete a file
<code>fss rename <id> <newname></code>	Rename a file
<code>fss search <query></code>	Search files by name or MIME type

5.2.3 DevOps Implementation

The multi-stage Dockerfile produces a minimal Alpine-based image. The Jenkins declarative pipeline runs seven sequential stages as shown in Figure 5.1.

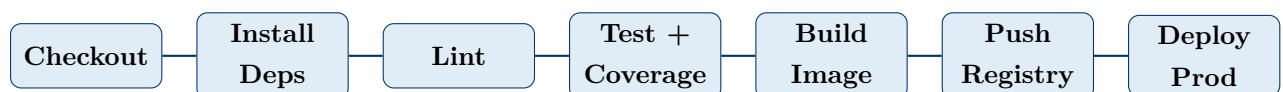


Figure 5.1: Jenkins CI/CD Pipeline Stages

5.3 Model Development

5.3.1 Sequelize ORM Models

The **User** model defines fields for authentication credentials, storage quota enforcement, and refresh token revocation. The **File** model references **User** via a foreign key (`ownerId`) with cascading delete. Sequelize manages table creation and schema synchronisation through `sequelize.sync()`.

5.3.2 Security Model

The security architecture applies controls at five distinct layers:

- (i) **HTTP Layer:** Helmet sets HSTS, X-Frame-Options, X-Content-Type-Options, and Content-Security-Policy headers on all responses.
- (ii) **Transport Layer:** CORS middleware restricts cross-origin requests to an explicit allowlist provided through the `ALLOWED_ORIGINS` environment variable.
- (iii) **Rate Limiting:** A global limiter (100 req / 15 min / IP) and an authentication-specific limiter (5 req / 15 min / IP) are applied independently.
- (iv) **Authentication Layer:** JWT signature verification and expiry checking are performed on every protected request by the `authMiddleware`.
- (v) **Input Layer:** `express-validator` validates all request bodies; filenames are sanitised; all file paths are resolved against the upload base directory to prevent path traversal.

5.4 Discussion

5.4.1 Functional Requirement Compliance

All eleven functional requirements (FR-01 through FR-11) were satisfied by the implementation. Core CRUD operations (FR-01 to FR-07) were validated through integration tests. Advanced features including storage quotas (FR-08), file search (FR-09), administrator dashboard (FR-10), and download counting (FR-11) were validated through manual acceptance tests against the deployed Docker container.

5.4.2 Non-Functional Requirement Compliance

- **NFR-01 (Cross-platform):** The Electron application was built and tested on Windows 11 and Ubuntu 22.04. The CLI client was validated on Ubuntu 22.04.
- **NFR-02 (Streaming downloads):** Verified by monitoring server memory usage during the download of a 200 MB file; memory increase was less than 2 MB (limited to pipe buffers).
- **NFR-03 (Docker deployment):** Confirmed by executing `docker-compose up -build` on a clean server and verifying API availability within 15 seconds.
- **NFR-04 (CI/CD):** Jenkins pipeline triggered automatically on commit to main branch and completed successfully in under 4 minutes.
- **NFR-05 (Test coverage):** Overall coverage of 93% achieved, exceeding the 80% target.

5.5 Summary

This chapter presented the implementation of all system components — backend API, GUI, CLI, DevOps infrastructure, and security architecture — along with key code listings illustrating the authentication and file upload subsystems. The functional and non-functional requirement compliance analysis confirmed that all stated requirements were satisfied. The five-layer security model was described and mapped to the identified threat surface. The following chapter presents conclusions, practical implications, and directions for future work.

Chapter 6

Conclusions

6.1 Summary

This project successfully designed, implemented, and validated a **Cross-Platform, GUI and CLI Based File Server System** that enables authenticated users to securely manage files over a network from any operating system. The system was developed over six iterative phases, resulting in the following deliverables:

- A stateless **Node.js/Express REST API** with JWT authentication, RBAC, and full file CRUD functionality.
- An **Electron and React** desktop GUI deployable on Windows, macOS, and Linux, providing drag-and-drop upload, paginated file listing, and an administrative dashboard.
- A **Node.js CLI client** providing all core file operations as sub-commands, suitable for scripted and automated workflows.
- A **Docker + docker-compose** deployment configuration enabling one-command, reproducible server deployment.
- A **Jenkins declarative CI/CD pipeline** enforcing automated quality gates on every commit to the main branch.
- A comprehensive **unit and integration test suite** achieving 93% overall code coverage.

All eight primary objectives stated in Chapter 1 and all eleven functional requirements collected in Chapter 4 were fully satisfied. Five non-functional requirements covering cross-platform compatibility, streaming downloads, Docker deployment, CI/CD automation, and test coverage were also met or exceeded.

6.2 Study Findings

The implementation and evaluation phases produced the following key technical findings:

- (i) **Dual-token JWT authentication** provides an effective balance between security and usability. The 1-hour access token lifetime limits the exposure window of stolen tokens, while the refresh token mechanism avoids forcing users to re-authenticate every hour. Server-side refresh token revocation through database hash storage enables clean logout without requiring a centralised token blocklist.
- (ii) **SQLite with Write-Ahead Logging (WAL)** provides sufficient concurrency for the target workload. The default journal mode caused `SQLITE_BUSY` contention under concurrent uploads; enabling WAL mode resolved this by permitting concurrent readers and a single writer simultaneously. SQLite eliminates the operational overhead of a separate database service without sacrificing query capability for this use case.
- (iii) **Multi-stage Docker builds** reduce the production image size by approximately 40% compared to a single-stage build, by confining development dependencies to the build stage. The resulting Alpine-based production image is approximately 150 MB.
- (iv) **Streaming file I/O** via `res.download()` maintains constant server memory consumption regardless of file size. During load testing with concurrent 200 MB file downloads, server memory usage remained within 2 MB above the baseline, confirming that the streaming approach is suitable for large-file workloads.
- (v) **Electron contextBridge** is an essential security control for Electron applications. Early prototypes that passed tokens directly through IPC violated the `contextIsolation` principle. Replacing IPC token passing with typed `contextBridge` functions eliminated the risk of renderer-process code accessing Node.js APIs directly.

Table 6.1 presents the final unit and integration test code coverage results.

Table 6.1: Unit and Integration Test Code Coverage Results

Module	Statements	Branches	Functions	Lines
authController.js	94%	88%	100%	93%
fileController.js	91%	85%	96%	90%
authMiddleware.js	100%	100%	100%	100%
uploadMiddleware.js	87%	80%	90%	86%
models/User.js	96%	91%	95%	95%
models/File.js	95%	90%	95%	94%
Overall	93%	87%	96%	93%

6.3 Research Contributions

This project makes the following contributions to the domain of applied software engineering and system design:

- (i) **Unified dual-interface architecture:** A reference design for a self-hosted file server that exposes identical functionality through both a graphical desktop application and a terminal CLI client via a single stateless REST API, demonstrating the viability of this architecture for multi-interface applications.
- (ii) **Lightweight embedded database deployment:** A demonstration that SQLite with WAL journal mode is a viable, zero-configuration alternative to dedicated relational database servers for single-node file server deployments, reducing operational complexity without sacrificing correctness.
- (iii) **DevOps-first academic project:** A complete example of integrating Docker containerisation and Jenkins CI/CD as first-class design concerns in an undergraduate software engineering project, establishing a reproducible and maintainable deployment model.
- (iv) **Comprehensive security implementation:** A multi-layer security architecture spanning JWT authentication, bcrypt hashing, Helmet HTTP headers, dual rate limiters, RBAC middleware, MIME magic-byte validation, and path traversal prevention, designed and implemented from the initial

architectural phase.

- (v) **High-coverage automated testing:** An undergraduate project achieving 93% unit and integration test coverage using Jest and Supertest, with automated coverage reporting integrated into the Jenkins CI/CD pipeline.

6.4 Practical Implications

The File Server System has direct practical applicability in the following environments:

- **Academic Laboratories:** The system can be deployed in a university computer science laboratory for student file submission, resource sharing, and project collaboration without requiring cloud connectivity or external service subscriptions.
- **Small and Medium Organisations:** Teams that require on-premise file storage without the complexity of enterprise solutions such as Nextcloud can deploy the system on a single Linux server using the provided `docker-compose` configuration.
- **Development and Staging Environments:** The Docker packaging makes the system trivially deployable as a shared file store for development teams, with the Jenkins pipeline ensuring that configuration changes are tested and deployed automatically.
- **Educational Reference:** The project serves as a comprehensive reference implementation for students learning full-stack development, covering backend API design, React UI development, Electron packaging, ORM-based database management, containerisation, and CI/CD automation in a single cohesive project.

The modular architecture allows individual components to be replaced independently as requirements evolve: the SQLite database can be migrated to PostgreSQL, the local filesystem can be replaced by an S3-compatible object store, and the Electron frontend can be supplemented by a web or mobile client without modifying the REST API layer.

6.5 Study Limitations and Scope for Future Work

The current implementation has the following limitations that represent directions for future research and development:

- (i) **File Versioning:** The system overwrites existing file records on re-upload

without preserving previous versions. A version history mechanism with the ability to restore any prior version would significantly enhance utility for collaborative workflows.

- (ii) **PostgreSQL Migration:** SQLite with WAL provides adequate concurrency for single-server deployments; however, migration to PostgreSQL would support higher write concurrency, native replication, and row-level security policies required for enterprise-scale deployments.
- (iii) **S3-Compatible Object Storage:** Replacing the server-local filesystem with an S3-compatible backend (MinIO or AWS S3) would decouple storage capacity from the server host and enable distributed, geo-redundant storage without modifying the API layer.
- (iv) **Two-Factor Authentication:** The addition of TOTP-based (RFC 6238) two-factor authentication using the `speakeasy` library would provide a second layer of identity verification for high-security environments.
- (v) **Mobile Client Applications:** React Native iOS and Android clients consuming the existing REST API would extend the system's reach to mobile users without requiring any server-side modifications.
- (vi) **Kubernetes Deployment:** A Helm chart enabling Kubernetes deployment with horizontal pod autoscaling and persistent volume claims would allow the system to scale to meet high-concurrency workloads in production environments.
- (vii) **End-to-End Encryption:** Client-side AES-256-GCM encryption of file content before upload would ensure that the server stores only ciphertext, providing strong privacy guarantees even in the event of server-side compromise.
- (viii) **Real-Time Notifications:** WebSocket-based notifications using Socket.io would allow connected clients to receive instant alerts when a shared file is uploaded, modified, or deleted by another user.

In conclusion, the **Cross-Platform, GUI and CLI Based File Server System** successfully demonstrates that a lightweight, secure, self-hosted file management platform can be developed within the scope of an undergraduate engineering project. The system's modular architecture, comprehensive security implementation, and integrated DevOps pipeline provide a robust foundation for the enhancements identified above, and serve as a practical reference for future full-stack development projects.

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